MSG No. 17.12 P_c222_1 [Type IV, orthorhombic]

Table 1: Wyckoff site: 2a, site symmetry: 22'2'

No.	position	mapping
1	$[0, 0, 0]$	[1,2,7,8]
2	$\left[0, 0, \frac{1}{2}\right]$	[3,4,5,6]

Table 2: Wyckoff site: 2b, site symmetry: 22'2'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,2,7,8]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[3,4,5,6]

Table 3: Wyckoff site: 2c, site symmetry: 22'2'

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	[1,2,7,8]
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[3,4,5,6]

Table 4: Wyckoff site: 2d, site symmetry: 2'22'

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	[1,3,6,8]
2	$\left[0, 0, \frac{3}{4}\right]$	[2,4,5,7]

Table 5: Wyckoff site: 2e, site symmetry: 22'2'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[1,2,7,8]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[3,4,5,6]

Table 6: Wyckoff site: 2f, site symmetry: 2'22'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[1,3,6,8]
2	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[2,4,5,7]

Table 7: Wyckoff site: 2g, site symmetry: 2'22'

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[1,3,6,8]

2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,4,5,7]`

Table 8: Wyckoff site: `2h`, site symmetry: `2'22'`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,3,6,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,4,5,7]`

Table 9: Wyckoff site: `4i`, site symmetry: `2..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,2]`
2	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[3,4]`
3	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[5,6]`
4	$[-x,\ 0,\ 0]$	`[7,8]`

Table 10: Wyckoff site: `4j`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,6]`
2	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[2,5]`
3	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[3,8]`
4	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[4,7]`

Table 11: Wyckoff site: `4k`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,2]`
2	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,4]`
3	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[5,6]`
4	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[7,8]`

Table 12: Wyckoff site: `4l`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,6]`
2	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,5]`
3	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[3,8]`
4	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[4,7]`

Table 13: Wyckoff site: `4m`, site symmetry: `.2'.`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,7]`
2	$[0,\ -y,\ 0]$	`[2,8]`
3	$\left[0,\ y,\ \frac{1}{2}\right]$	`[3,5]`
4	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[4,6]`

Table 14: Wyckoff site: `4n`, site symmetry: `.2.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,3]`
2	$\left[0,\ -y,\ \frac{3}{4}\right]$	`[2,4]`
3	$\left[0,\ y,\ \frac{3}{4}\right]$	`[5,7]`
4	$\left[0,\ -y,\ \frac{1}{4}\right]$	`[6,8]`

Table 15: Wyckoff site: `4o`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ 0\right]$	`[1,7]`
2	$\left[\frac{1}{2},\ -y,\ 0\right]$	`[2,8]`
3	$\left[\frac{1}{2},\ y,\ \frac{1}{2}\right]$	`[3,5]`
4	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	`[4,6]`

Table 16: Wyckoff site: `4p`, site symmetry: `.2.`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ \frac{1}{4}\right]$	`[1,3]`
2	$\left[\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	`[2,4]`
3	$\left[\frac{1}{2},\ y,\ \frac{3}{4}\right]$	`[5,7]`
4	$\left[\frac{1}{2},\ -y,\ \frac{1}{4}\right]$	`[6,8]`

Table 17: Wyckoff site: `4q`, site symmetry: `..2'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,8]`
2	$[0,\ 0,\ -z]$	`[2,7]`
3	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[3,6]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[4,5]`

Table 18: Wyckoff site: `4r`, site symmetry: `..2'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,8]`

continued ...

Table 18

No.	position	mapping
2	$\left[\frac{1}{2}, 0, -z\right]$	[2,7]
3	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[3,6]
4	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	[4,5]

Table 19: Wyckoff site: 4s, site symmetry: ..2'

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,8]
2	$\left[0, \frac{1}{2}, -z\right]$	[2,7]
3	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[3,6]
4	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	[4,5]

Table 20: Wyckoff site: 4t, site symmetry: ..2'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[1,8]
2	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[2,7]
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	[3,6]
4	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	[4,5]

Table 21: Wyckoff site: 8u, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[x, -y, -z]$	[2]
3	$\left[-x, y, \frac{1}{2} - z\right]$	[3]
4	$\left[-x, -y, z + \frac{1}{2}\right]$	[4]
5	$\left[x, y, z + \frac{1}{2}\right]$	[5]
6	$\left[x, -y, \frac{1}{2} - z\right]$	[6]
7	$[-x, y, -z]$	[7]
8	$[-x, -y, z]$	[8]